

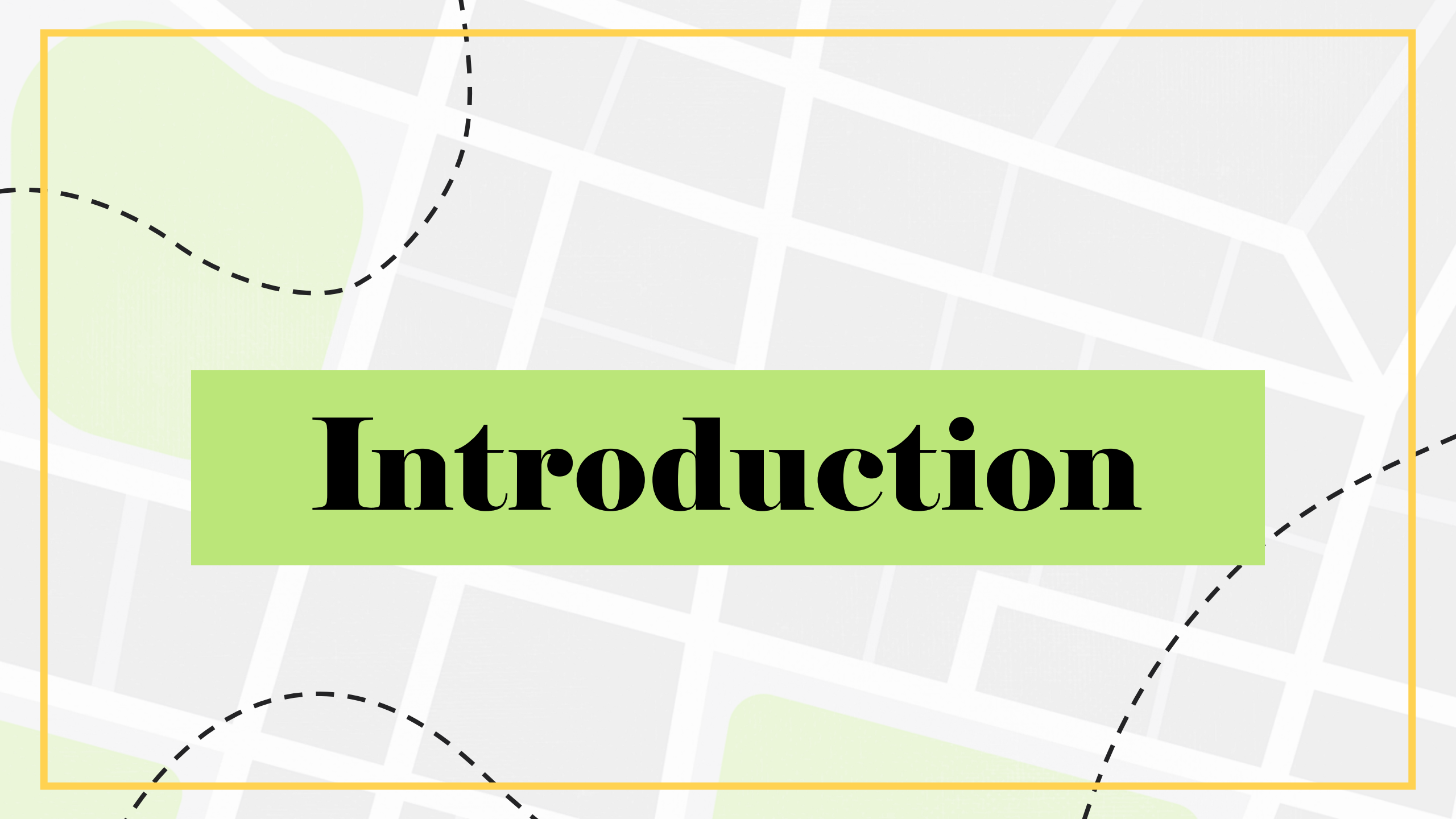


QMAP

**Elijah Alston
Madison Hamelin
Asifa Iqbal**

Table of Contents

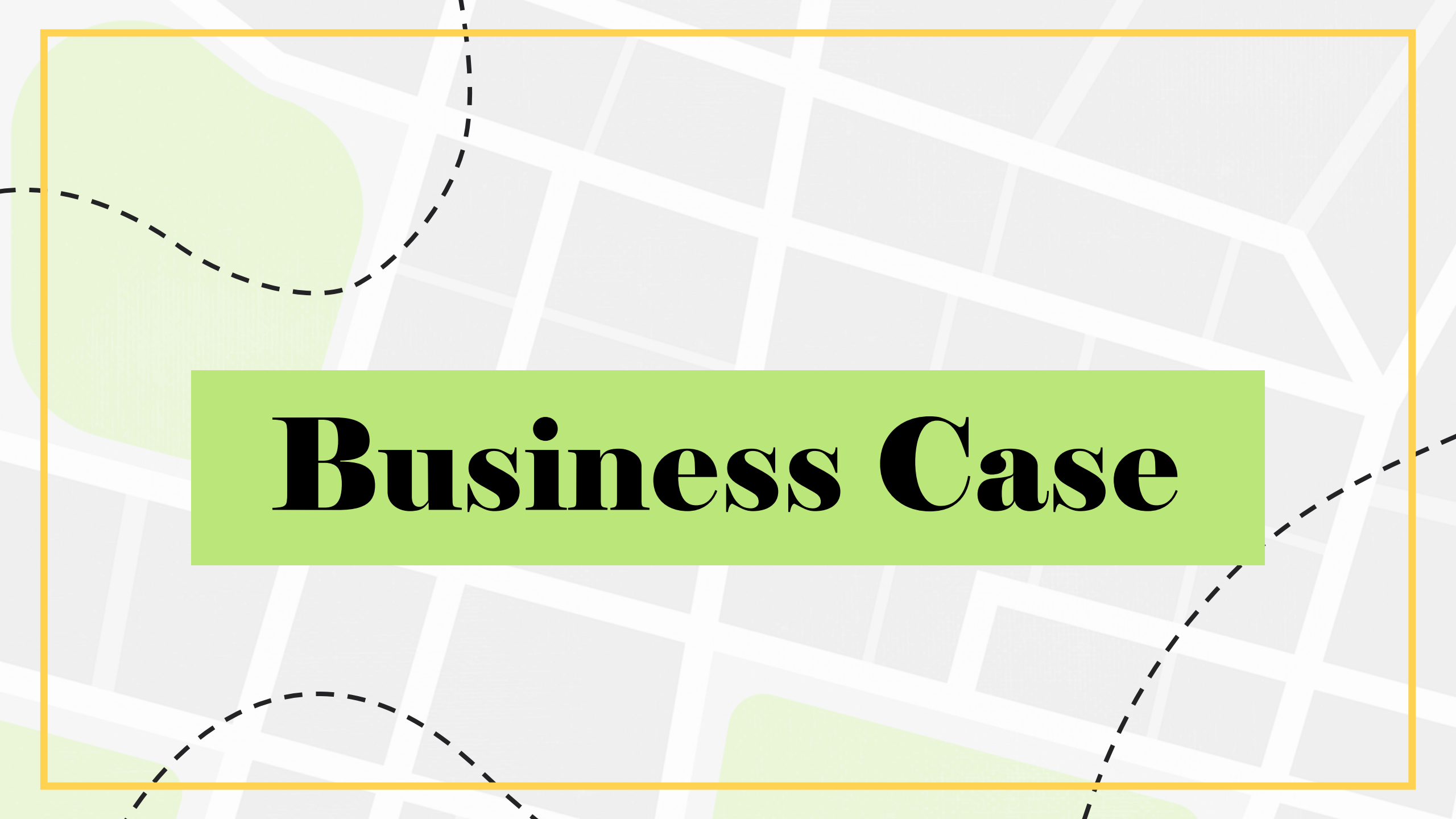
Introduction
Business Case
Project Timeline
User & Stakeholder Analysis
User Requirements
Functional Requirements
Nonfunctional Requirements
Process Models (DFD)
Process Description
Data Model
Data Dictionary
User Interfaces
References

The background features a stylized map with light gray rectangular blocks representing buildings and white lines representing streets. A large, irregular green shape is on the left side. A solid orange rectangular border frames the entire content. A dashed black line meanders across the background, passing through the green shape and the orange border.

Introduction

Introduction

- General navigation applications are ineffective for indoor environments leading to frustration navigating Mount Carmel campus
- Q-Map, a student developed navigation application tailored specifically for Mount Carmel campus

The background features a stylized map with light gray rectangular blocks representing buildings and white lines representing streets. A solid green rectangle is centered horizontally, containing the text 'Business Case'. A dashed black line meanders across the image, passing behind the green rectangle. The entire composition is framed by a thin orange border.

Business Case

SWOT Analysis

Strengths

- Real-time updates on locations like classroom, bathrooms, and water fountains
- Syncs up with calendar and campus maps
 - Bathroom maintenance updates

Weaknesses

- Dependent on the speed and reliability of data updates
- Adaptation and maintenance challenges
 - Poor internet or GPS coverage

Opportunities

- Enhance student life
- Differentiate itself
- Quinnipiac branding
- In-app advertisements

Threats

- Established competitors
- Reporting campus closures and maintenance

Cost Benefit Analysis

Basic App Cost

\$10,000- \$50,000

UI/UX Design

\$5,000-\$20,000

AI/ML

\$25,000-\$50,000

Recurring Cost

\$10,000--\$30,000

Feasibility Assessment

Technical

- Resourceful tool to help users find any destination on Mount Carmel campus
- Uses previously collected data on rooms, bathrooms and offices

Economic

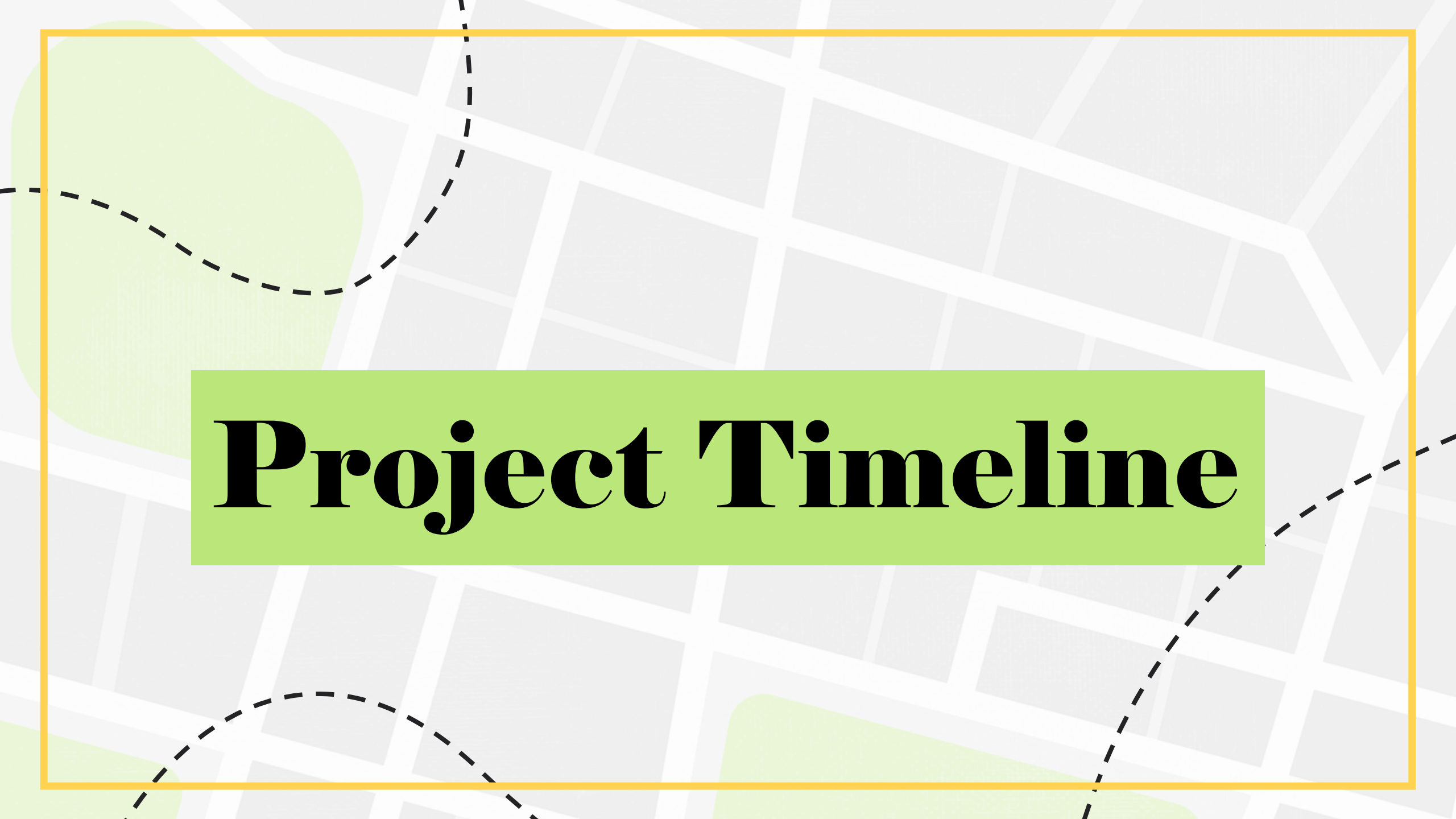
- Funded by Quinnipiac University
- Success profitability may lead to expansion

Legal/Contractual

- Must accept app's terms and conditions
- Sharing is only required when the app is open

Operational

- Very easy to navigate through to locate classes during the first few weeks of school
- GPS- like system tailored to Mount Carmel campus

The background features a light gray grid of squares, some of which are white, creating a map-like pattern. A large, irregular light green shape is on the left side. A dashed black line curves across the top and bottom of the image. A solid orange border frames the entire content.

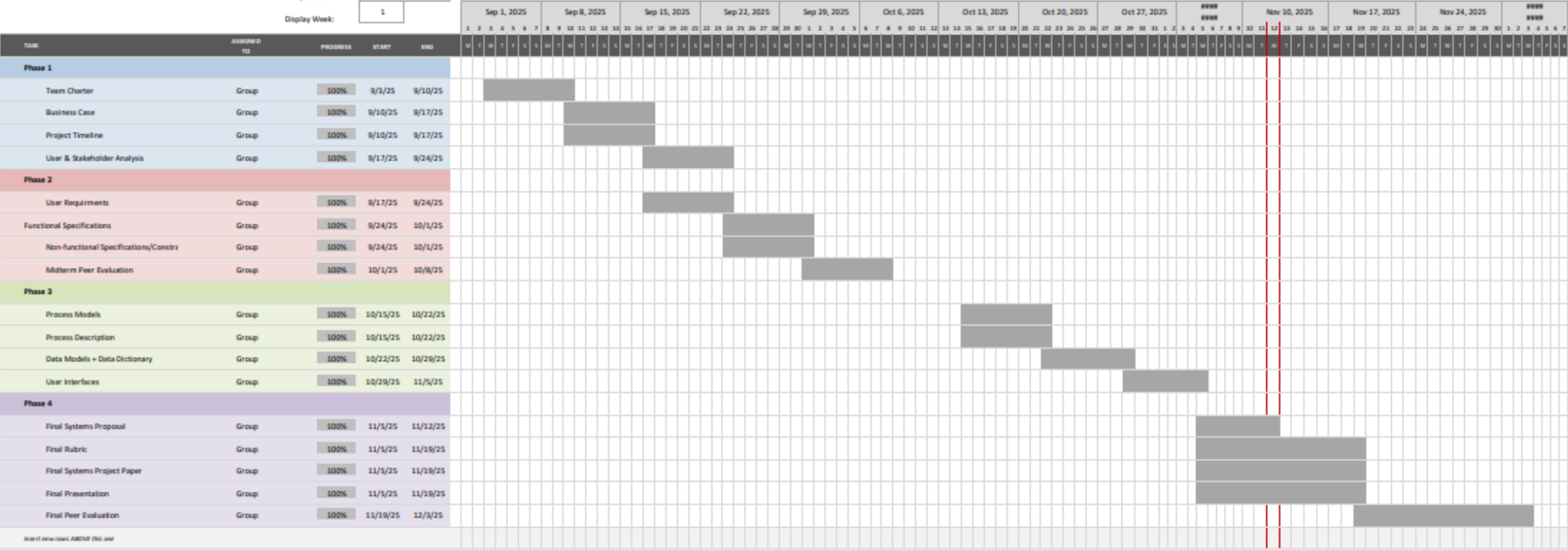
Project Timeline

Project Timeline

Q-Map

Team S:
Madison Hamelin, Iqbal Auliz, Alton Elijah

Project Start:
Display Week:



Want new rows ABOVE this one

The background features a stylized map with light gray rectangular blocks representing buildings or land parcels, separated by white lines. There are several irregular green shapes scattered across the map, possibly representing parks or water bodies. A dashed black line meanders across the map, starting from the left edge, curving upwards, then downwards, and finally extending towards the bottom right corner. A solid orange rectangular border frames the entire image.

User & Stakeholder Analysis

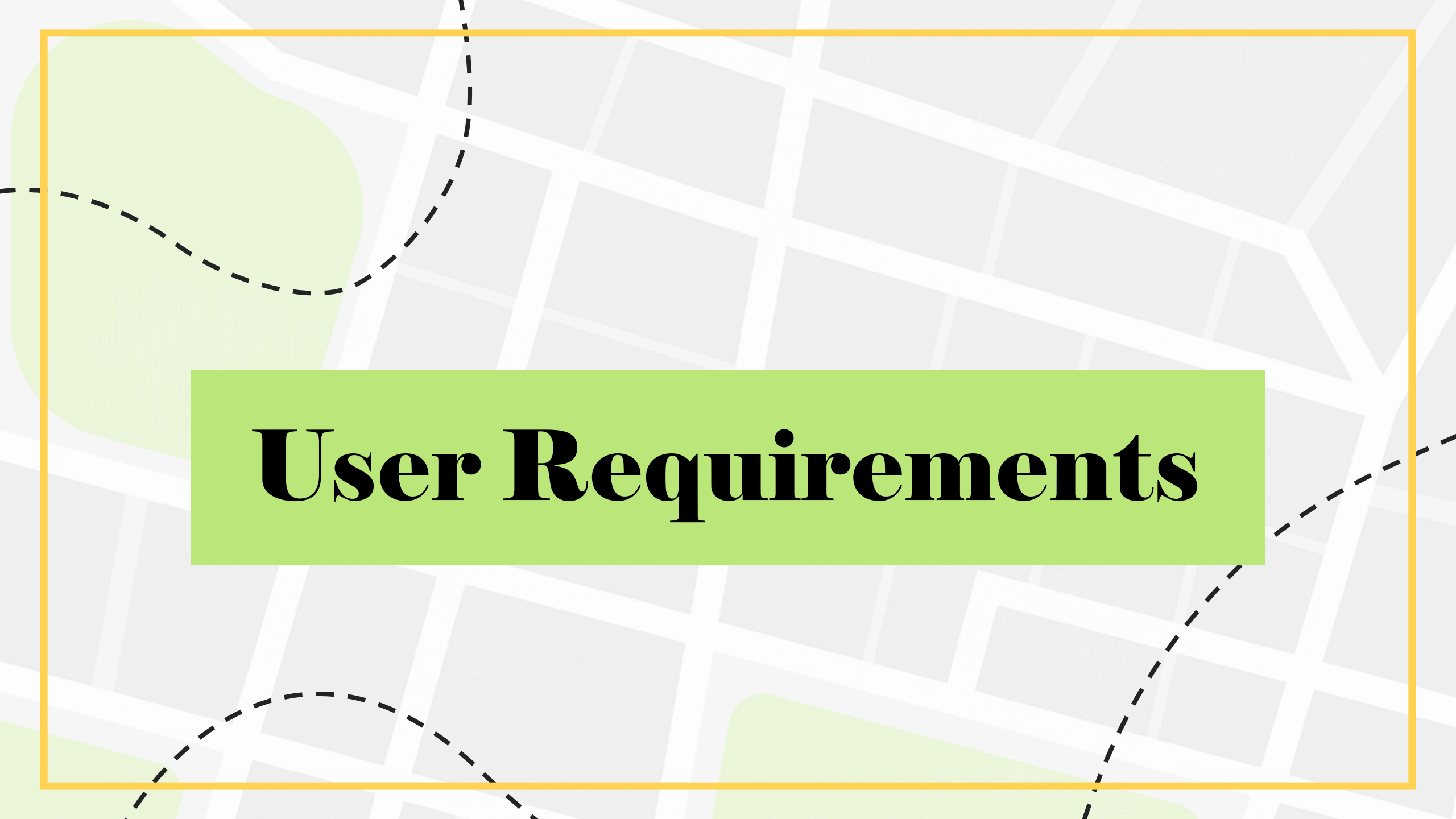
User & Stakeholder Analysis

Purpose of our System:

- Provide turn-by-turn campus navigation from students, faculty, and visitors' current location to their destination
- Display real-time status of campus facilities
- Offer travel-time estimates and leave-time notifications
- Show an interactive campus map with searchable locations

Stakeholders outside primary users:

- QU IT department, upper management and administration, etc.

The background features a stylized map with light gray rectangular blocks representing buildings and white lines representing streets. A large, solid green rectangle is centered horizontally, containing the text 'User Requirements'. Dashed black lines are scattered across the map, some forming arcs and others as straight segments. A thin orange border frames the entire image.

User Requirements

User Requirements

Students:

- Navigate campus with GPS and reminders
- Receive real-time updates on closures, delays, and room availability
- Use location services for accurate tracking and navigation

Faculty:

- Sync with lecture and meeting schedules
- Efficient campus navigation
- Alerts for closures, maintenance, or cancellations

Visitors:

- Directions and information for key campus locations
- See open and accessible areas
- View building hours and accessible paths

IT Department:

- Integrate with campus databases for locations and travel times
- Protect user data and comply with privacy policies
- Support search and system performance

Upper management and administration:

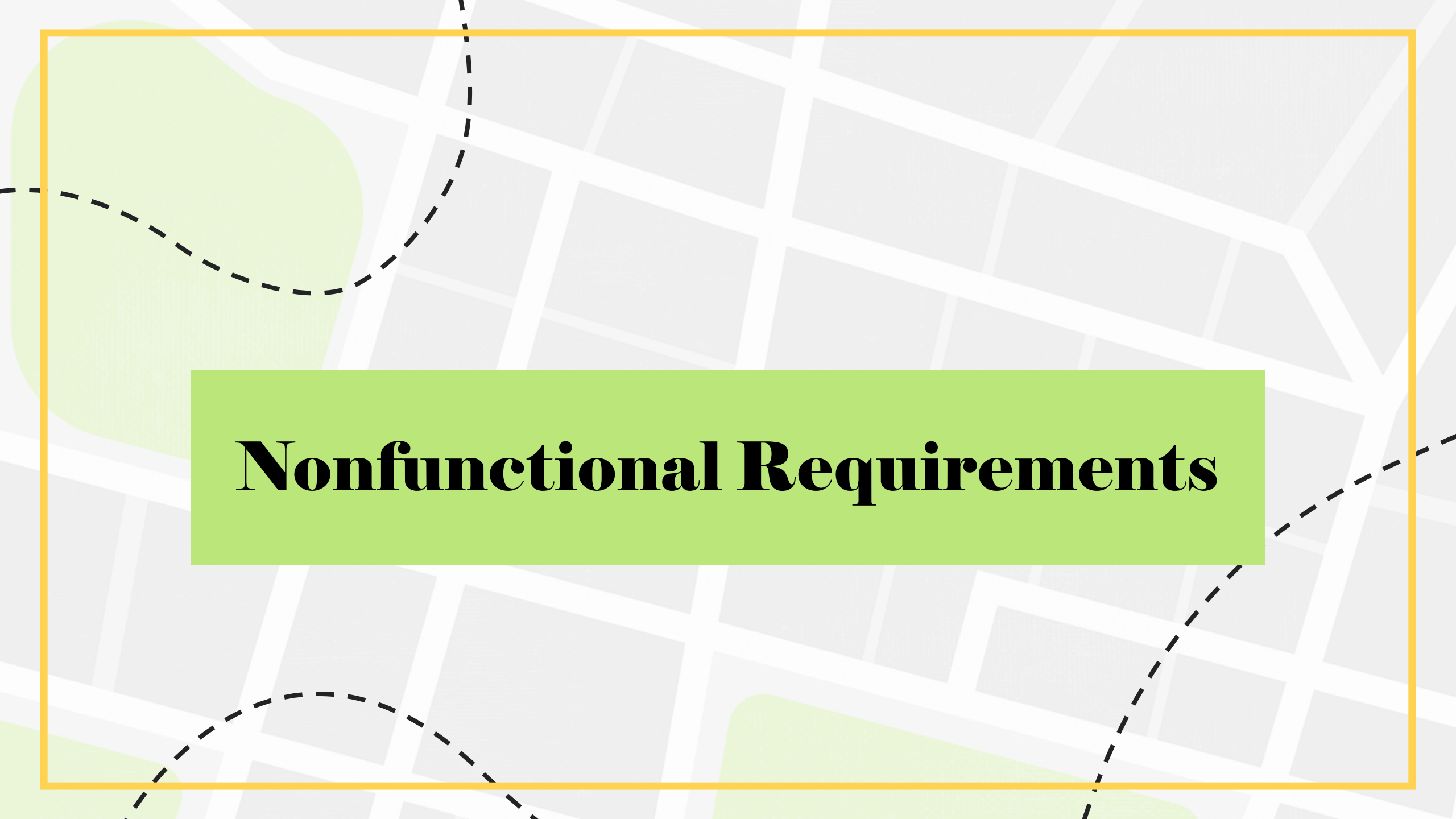
- Maintain app within budget
- Access usage analytics and performance reports
- Ensure app aligns with university mission and student success



Functional Requirements

Functional Requirements

1. User login interface
2. Interactive campus map
3. Search bar functionality
4. Personalized user information
5. Help and support interface
6. Alert and notification screen



Nonfunctional Requirements

Nonfunctional Requirements

Operational:

- User-friendly interface with intuitive navigation
- Integrate with QU's existing real-time alert system
- Shall remain operational throughout the academic year

Performance:

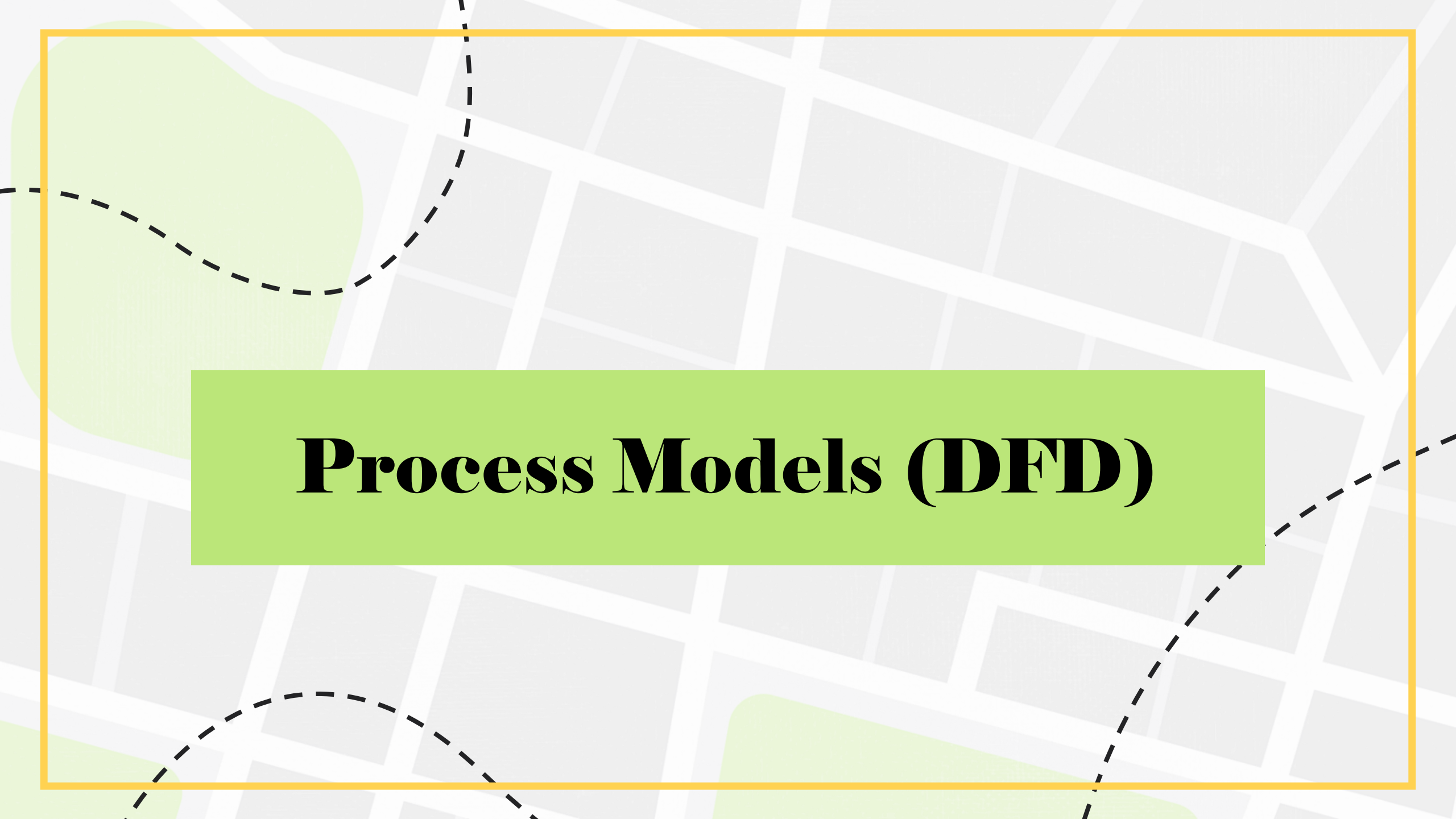
- Screens and alerts displayed immediately
- GPS and travel times accurate
- Supports thousands of simultaneous users

Security:

- Location visible only to user
- Store and protect confidential information
- Sign-in required for access

Cultural & Political:

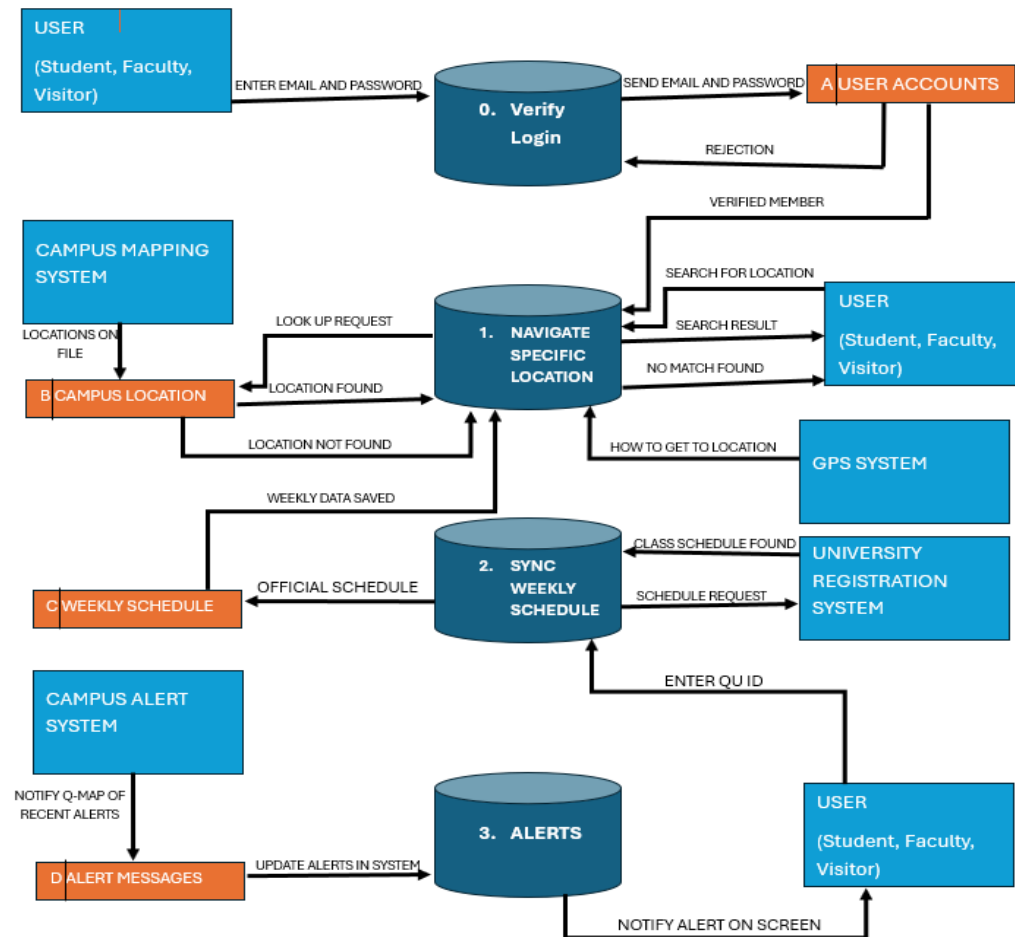
- Must comply with Quinnipiac University regulations and policies
- Must comply with local, state, and federal law and regulations



Process Models (DFD)

Process Models (DFD)

Q-MAP PROCESS MODELS (DFD)



The background features a stylized map with light gray rectangular blocks representing buildings and white lines representing streets. A large, solid green rectangle is centered horizontally, containing the text. Dashed black lines curve across the map, and a thin orange border frames the entire scene.

Process Description

Process Description

Process 0: Verify Login

DO RETRIEVE login input from the user, SEND credentials to the User Accounts data store, and VALIDATE them by checking for a match and active status. IF the credentials are verified, SET the session as logged in and RETURN a login confirmation. OTHERWISE, RETURN an error message and prompt the user to retry. UNTIL End-of-File.

Process 1: Navigate Specific Location

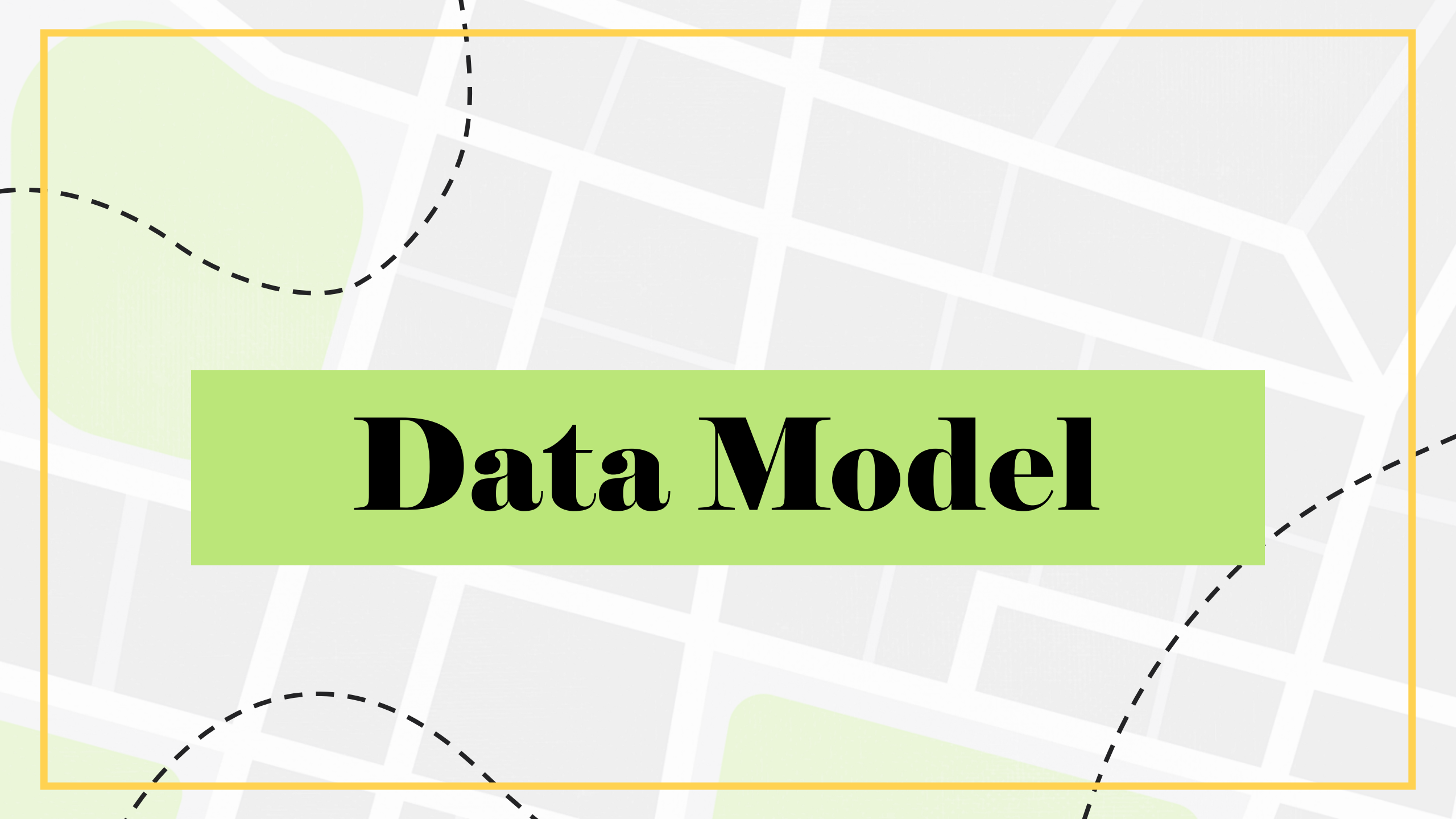
DO RETRIEVE the user's search request for a building, room, or point of interest, SEND a lookup request to the Campus Mapping System, and ACCESS the Campus Location data store for matching location details. IF a location is found, REQUEST route information from the GPS System, COMPUTE directions, and DISPLAY the map and navigation steps. OTHERWISE, RETURN a "No match found" message to the user. UNTIL End-of-File.

Process 2: Sync Weekly Schedule

DO RETRIEVE the user's sync request using their QU ID or portal credentials, SEND a schedule request to the University Registration System, and RECEIVE class schedule data. STORE the official schedule information in the Weekly Schedule data store and CONFIRM the sync success to the user while making the schedule available for navigation features. UNTIL End-of-File.

Process 3: Alerts

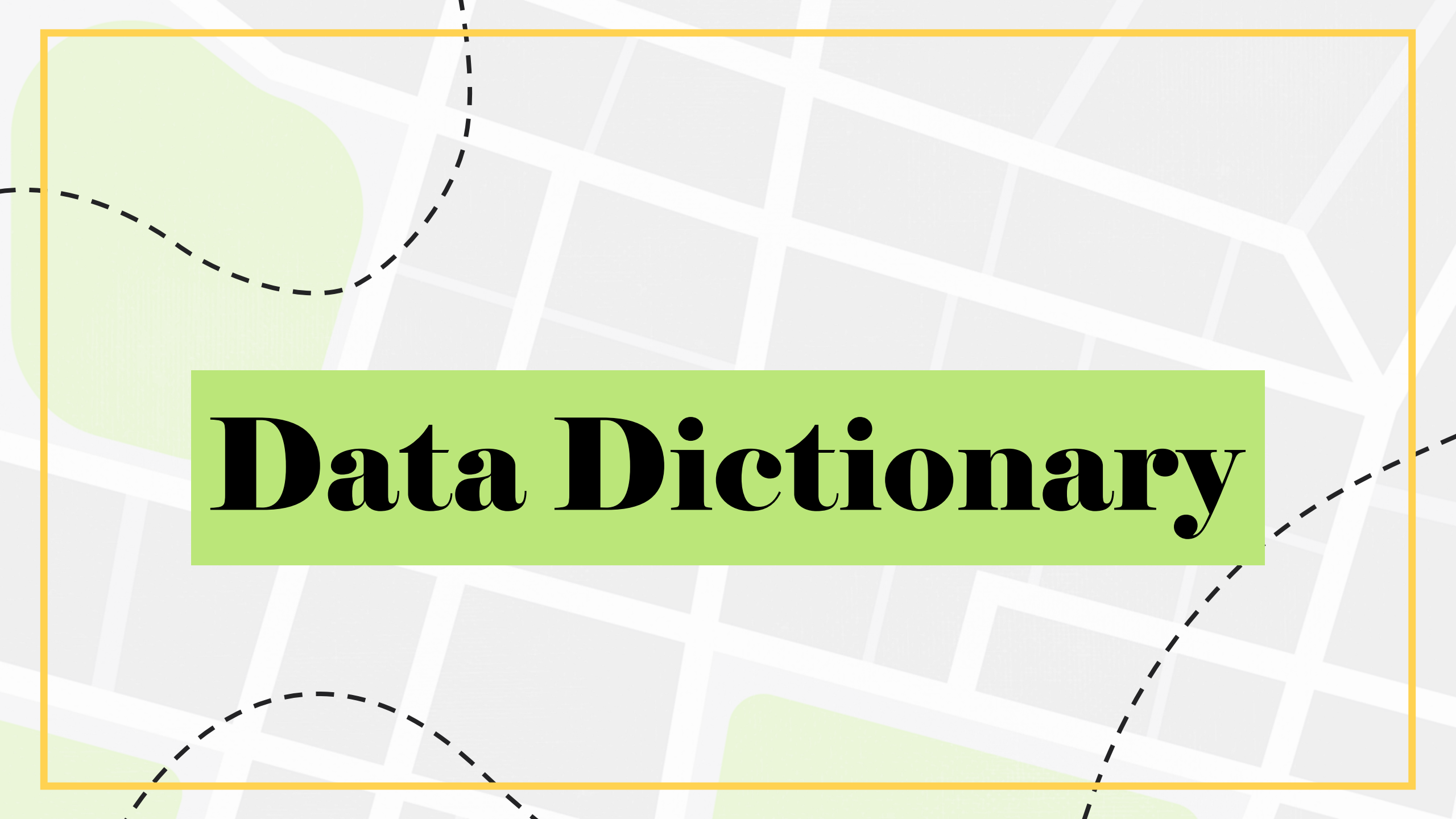
DO RECEIVE alert data from the Campus Alert System, UPDATE the Alert Messages data store with the new alert and its status, FILTER the alert based on user role, location, and relevance, and SEND and DISPLAY the notification on the user's device. UNTIL End-of-File.

The background features a stylized map with light gray rectangular blocks representing buildings and white lines representing streets. A large, solid green rectangle is centered horizontally, containing the text 'Data Model'. Several dashed black lines are overlaid on the map, forming curved paths that suggest movement or data flow. The entire composition is framed by a thin orange border.

Data Model

Data Model

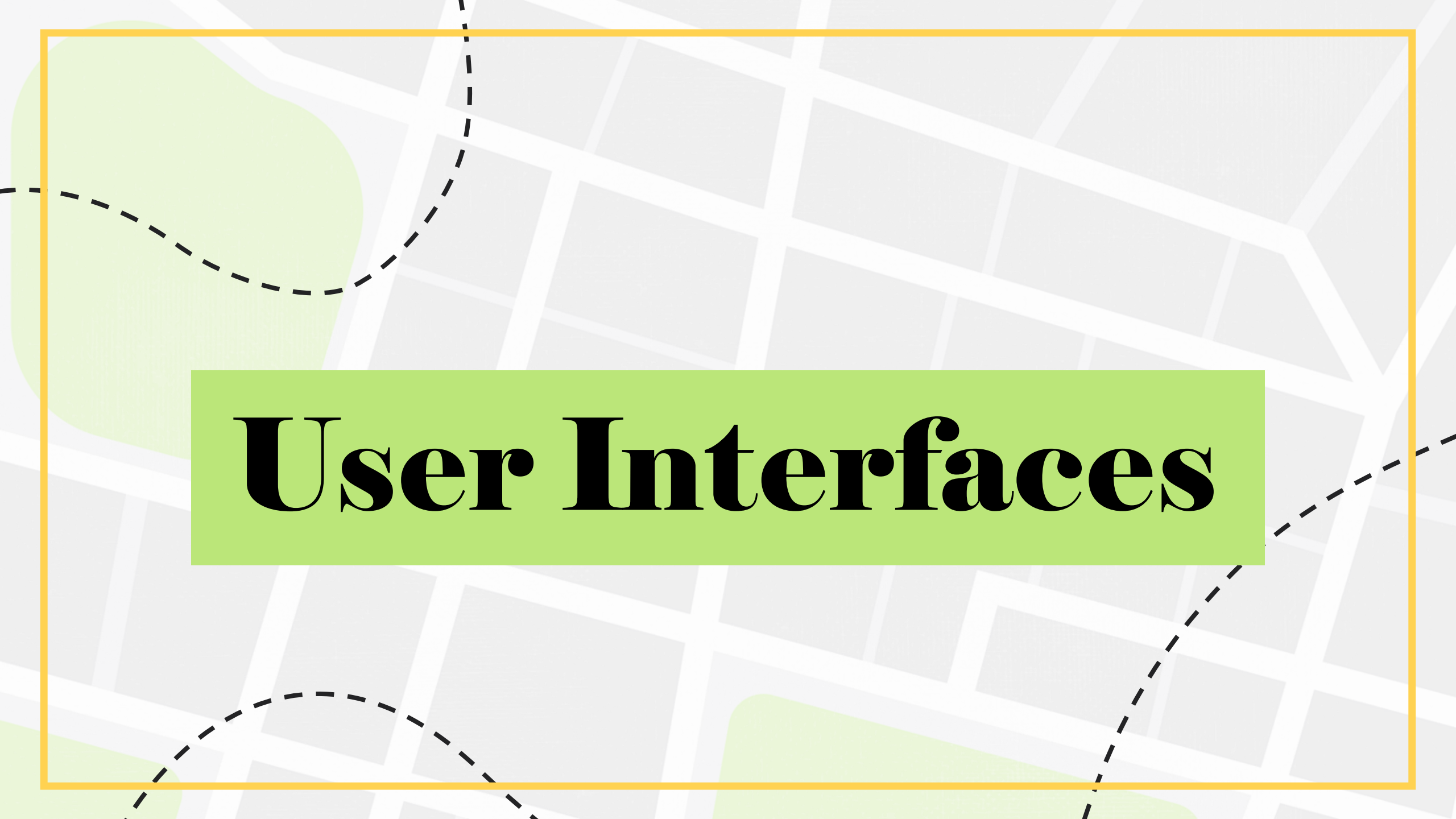
Data Store Name	Composition
Alert Messages	Affected_Area Alert_DateTime Alert_ID (pk) Alert_Message Alert_Status Alert_Type
Campus Location	Building_Name Building_Type Floor_Number Last_Updated Latitude Location_ID (pk) Longitude Operating_Hours Room_Number
User Accounts	Account_Status Last_Login QU_ID (pk) User_Email User_FName User_LName User_Role
Weekly Schedule	Building_Name Course_Days Course_Name Course_Time Instructor_Name QU_ID (fk) Room_Number

The background features a stylized map with light gray rectangular blocks representing buildings and white lines for streets. A solid green rectangle is centered horizontally, containing the text 'Data Dictionary'. Dashed black lines curve across the map, and a thin orange border frames the entire scene.

Data Dictionary

Data Dictionary

Data Element Name	Description	Type	Length
Account_Status	Current status of the user account (Active, Disabled, Pending)	Text	20
Affected_Area	Area or region impacted by the alert	Text	60
Alert_DateTime	When the alert was issued	Alphanumeric	25
Alert_ID (pk)	Unique identifier for each alert record	Number	10
Alert_Message	Content of the alert displayed to users	Text	100
Alert_Status	Current status of the alert (Active, Cleared)	Text	20
Alert_Type	Type of alert (Weather, Safety, Emergency)	Text	40
Building_Name	Full name of the building	Text	50
Building_Type	Category of building (Academic, Residential, Dining, etc.)	Text	40
Course_Days	Days of the week the class meets	Text	15
Course_Name	Title or name of the course	Text	60
Course_Time	Time interval when the class meets	Text	20
Floor_Number	Floor where the room is located	Number	2
Instructor_Name	Name of the course instructor	Text	50
Last_Login	Date/time the user last accessed the system	DateTime	—
Last_Updated	Date/time the building/location data was refreshed	DateTime	—
Latitude	GPS latitude coordinate	Decimal	6
Location_ID (pk)	Unique identifier for each location on campus	Number	10
Longitude	GPS longitude coordinate	Decimal	6
Operating_Hours	Hours during which the building or room is open	Text	50
QU_ID (fk/pk)	Unique identifier for an individual user	Number	7
Room_Number	Specific room identifier	Text	10
User_Email	User's registered email address	Text	60
User_FName	The first name of an individual user	Text	40
User_LName	The last name of an individual user	Text	40
User_Role	Indicates user type (Student, Faculty, Visitor)	Text	20

The background features a stylized map with light gray rectangular blocks representing buildings and white lines representing streets. A solid green rectangle is centered horizontally, containing the text 'User Interfaces'. A dashed black line curves across the image, passing behind the green rectangle. The entire composition is framed by a thin orange border.

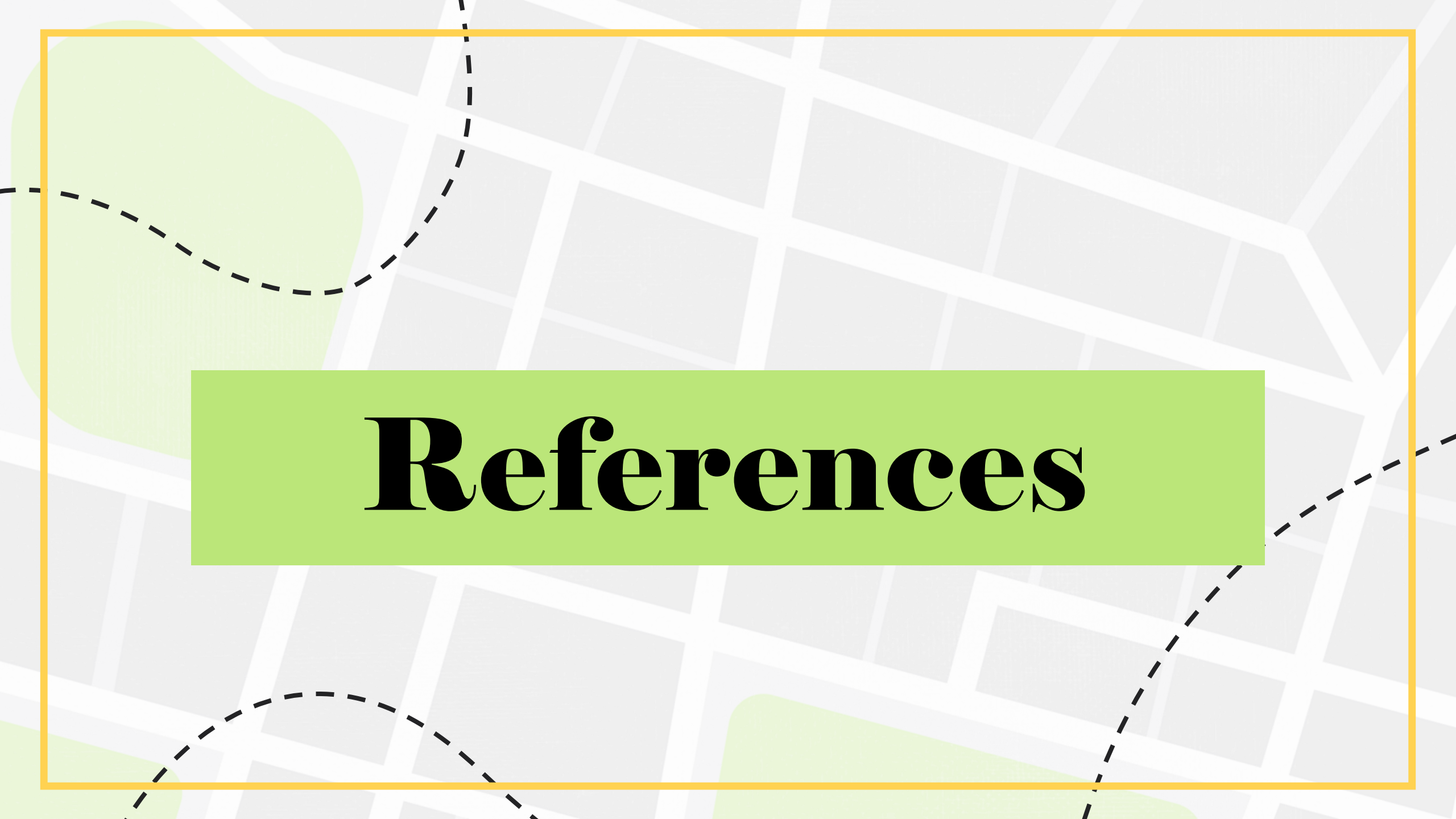
User Interfaces

User Interfaces

Click [HERE](#) to access the Marvel prototype of our app Q-MAP!

Note that we have only prototyped our student side and the faculty and visitor side would be adjusted accordingly!



The background features a stylized map with light gray rectangular blocks and white lines representing streets. A solid green rectangle is centered horizontally, containing the word 'References' in a bold, black, serif font. A dashed black line meanders across the slide, passing behind the green rectangle. The entire composition is framed by a thin orange border.

References

References

“Q-Map.” *Marvel*, marvelapp.com/prototype/1cai58ej. Accessed 18 Nov. 2025.